

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460328.0024 +/- 0.0028 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.15 +/- 0.012
Transit depth (Rp/Rs)^2	2.26 +/- 0.36 [%]
Orbital Inclination (inc)	84.6 +/- 0.3
Airmass coefficient 1 (a1)	1.1681 +/- 0.0064
Airmass coefficient 2 (a2)	-0.1088 +/- 0.0024
Scatter in the residuals of the lightcurve fit is	0.55 %
Best Comparison Star	None
Optimal Aperture	3.64
Optimal Annulus	7.17
Transit Duration (day)	0.0908 +/- 0.0079

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.15 ± 0.012 vs 0.1276 (deviation 1.61σ)
Duration fit vs published	0.0908 d vs 0.1179 d (deviation 2.95σ)
Mid-time O–C	24.1 min
Depth SNR	10.7
Residual scatter	0.55 %

Light curve

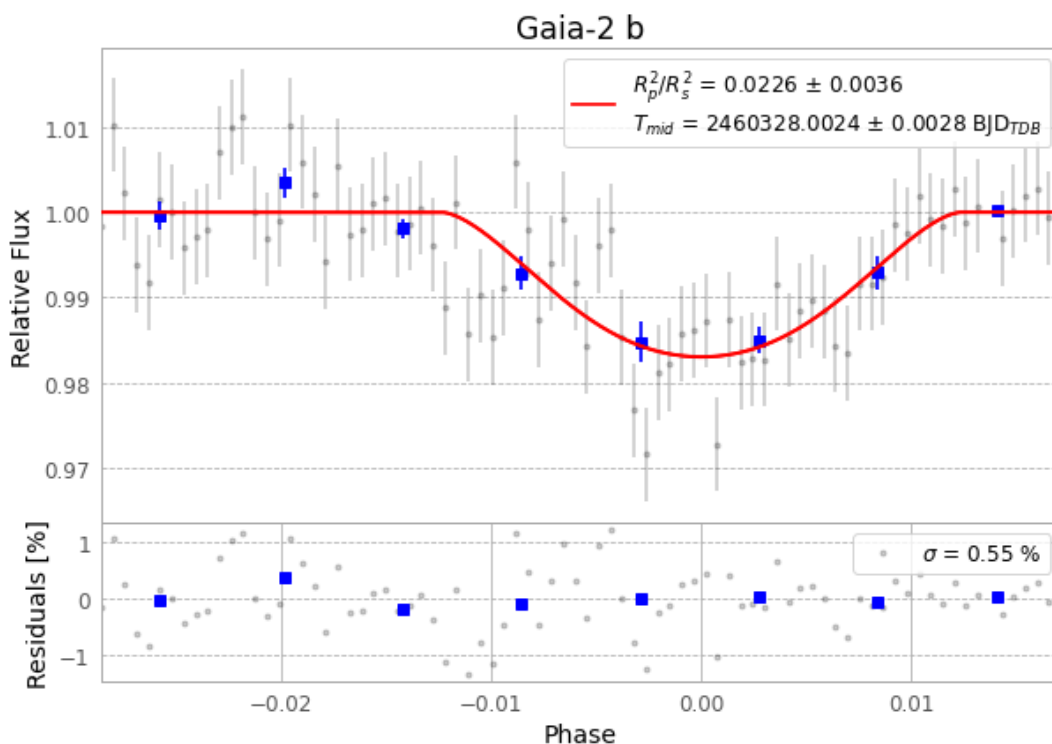


Figure 1 — FinalLightCurve_Gaia-2 b_2024-01-18.png (SHA-256 5f099c93fa76aa95...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [303, 221], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2d3b83dbf66bbe3b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

82 FITS frames (54 MB), Gaia-2240118092415.FITS → Gaia-2240118132715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Gaia-2-240118.log (1990055dfa2e...), FinalLightCurve_Gaia-2 b_2024-01-18.png (5f099c93fa76...), FinalLightCurve_Gaia-2 b_2024-01-18.csv (73fea307d373...)

Compute resources

Wall time 117.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-16 22:48Z.